Modern Physics Hw 1

Contents

- Question 1
- Question 1 - Answer
- Question 2
- Question 2 - Answer
- Question 3
- Question 3 - Answer


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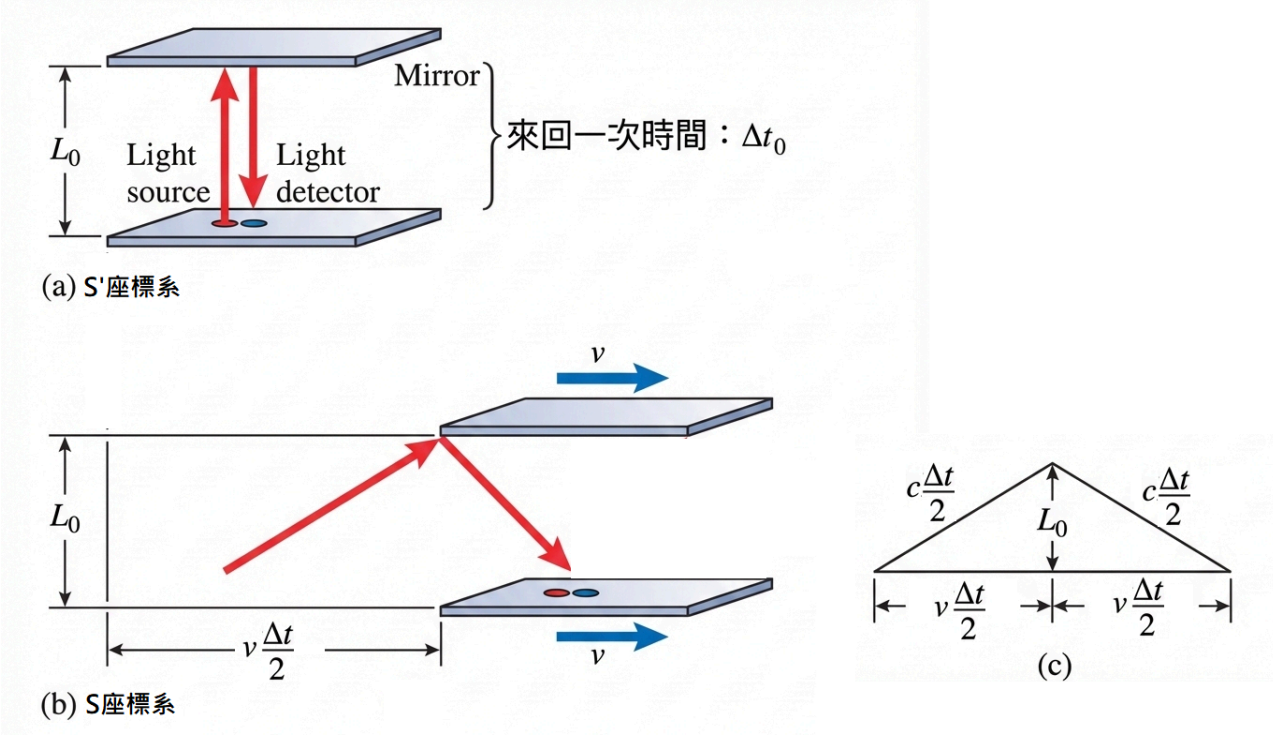
Question 1

Prove time dilation

$$t = \frac{2L_0/c}{\sqrt{1-v^2/c^2}} = \frac{t_0}{\sqrt{1-v^2/c^2}}$$

Question 1 - Answer

雖然題目是寫 t 但是我覺得寫 Δt 更加精準



假設飛船上有一個光鐘，光子在相距 L_0 的兩面鏡子間上下反射。

- **在飛船上 (S' 座標系)：** 觀察者看到光子直上直下，來回一次所需的時間為原時 Δt_0。

$$\frac{\Delta t_0}{2} = \frac{L_0}{c}$$

- **在地球上 (S 座標系)：** 觀察者看到飛船以速度 v 前進，因此光子走的是斜線路徑、設地球觀察者測量到的時間為 Δt，根據畢氏定理：

$$\left(c\frac{\Delta t}{2}\right)^2 = L_0^2 + \left(v\frac{\Delta t}{2}\right)^2$$

將 $L_0 = c\frac{\Delta t_0}{2}$ 代入：

$$
\begin{aligned}
c^2\left(\frac{\Delta t}{2}\right)^2 &= \left(c\frac{\Delta t_0}{2}\right)^2 + v^2\left(\frac{\Delta t}{2}\right)^2 \\
c^2(\Delta t)^2 &= c^2(\Delta t_0)^2 + v^2(\Delta t)^2 \\
(\Delta t)^2(c^2 - v^2) &= c^2(\Delta t_0)^2 \\
(\Delta t)^2 &= \frac{c^2}{c^2 - v^2}(\Delta t_0)^2 \\
(\Delta t)^2 &= \frac{1}{1 - v^2/c^2}(\Delta t_0)^2 \\
\Delta t &= \frac{1}{\sqrt{1 - v^2/c^2}}\Delta t_0 \\
\Delta t &= \frac{1}{\sqrt{1 - v^2/c^2}}\frac{2L_0}{c}
\end{aligned}
$$

核心概念： 靜止的觀察者會測量到，相對於他運動的時鐘，其滴答速度比他身旁靜止的時鐘還要慢。

Question 2

Prove that when v is much smaller than c , the kinetic energy reduces to the classical definition

$$KE = \gamma mc^2 - mc^2 = (\gamma - 1)mc^2$$

Question 2 - Answer

已知與假設

- 【已知 1】古典KE公式:

$$KE_{\text{classical}} = \frac{1}{2}mv^2$$

- 【已知 2】Lorentz factor γ:

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

- 【已知 3】數學近似-泰勒級數1階展開:

$$(1+\varepsilon)^n \approx 1 + n\varepsilon$$

- 【已知 4】數學近似:

$$\frac{a}{b+\varepsilon} \approx \frac{a}{b}$$

proof

$$\begin{aligned}
KE &= (\gamma - 1)mc^2 \\
&\overset{\text{已知 2}}{=} (\frac{1}{\sqrt{1-v^2/c^2}} - 1)mc^2 \\
&= \frac{1-\sqrt{1-v^2/c^2}}{\sqrt{1-v^2/c^2}}mc^2 \\
&\overset{\text{已知 3}}{=} \frac{1-(1-\frac{1}{2}v^2/c^2)}{(1-\frac{1}{2}v^2/c^2)}mc^2 \\
&= \frac{\frac{1}{2}v^2/c^2}{(1-\frac{1}{2}v^2/c^2)}mc^2 \\
&\overset{\text{已知 4}}{=} \frac{1}{2}v^2/c^2mc^2 \\
&= \frac{1}{2}v^2m \overset{\text{已知 1}}{=} KE_{\text{classical}}
\end{aligned}$$

Question 3

Prove equation:

$$E^2 = (mc^2)^2 + p^2c^2$$

Question 3 - Answer

我們希望找出不包含速度變數 v 的能量與動量關係。

已知與假設

- 【已知 1】物體總能量:

$$E = \gamma m_0 c^2$$

- 【已知 2】物體動量:

$$p = \gamma m_0 v$$

- 【已知 3】Lorentz factor γ:

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

proof

將總能量和動量平方後相減：

$$\begin{aligned}
E^2 - (pc)^2 &= (\gamma m_0 c^2)^2 - (\gamma m_0 v c)^2 \\
&= \gamma^2 m_0^2 c^4 - \gamma^2 m_0^2 v^2 c^2 \\
&= \gamma^2 m_0^2 c^4 \left(1 - \frac{v^2}{c^2}\right) \\
&= \gamma^2 m_0^2 c^4 \gamma^{-2} \\
&= m_0^2 c^4 \\
E^2 &= (m_0 c^2)^2 + (pc)^2
\end{aligned}$$